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Electrical connector and connector device

Abstract

An electrical connector is attached to a wiring substrate and mated with a counterpart connector connected to a signal transmission medium. A shell of the electrical connector has a pair of wall parts that contact a ground member on the counterpart connector. One wall part distant from the signal transmission medium includes a first side plate that has one end attached to a ground conductive path of the wiring substrate and extends in a direction away from the wiring substrate, a joining part that has one end joined to another end of the first side plate, and a second side plate that has one end joined to another end of the joining part and extends in a direction closer to the wiring substrate. The second side plate has a contact piece that extends in a direction away from the wiring substrate and elastically contacts the ground member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of and claims benefit under 35 U.S.C. § 120 to U.S. application Ser. No. 17/209,335, filed Mar. 23, 2021, which is a continuation of and claims benefit under 35 U.S.C. § 120 to U.S. application Ser. No. 16/720,967, filed Dec. 19, 2019, which is based upon and claims the benefit of priority under 35 U.S.C. § 119 from Japanese Patent Application No. 2018-248235, filed Dec. 28, 2018, the entire contents of each of which are herein incorporated by reference.

BACKGROUND OF THE INVENTION
1. Field of the Invention
(1) A disclosed embodiment relates to an electrical connector and a connector device.
2. Description of the Related Art
(2) Conventionally, a connector device for electrically connecting a signal transmission medium such as a coaxial cable, a flexible printed circuit (FPC), or a flexible flat cable (FFC) to a wiring substrate has been widely used in an electronic instrument. Such a connector device includes a first connector that is connected to a terminal of a signal transmission medium and a second connector that is connected to a wiring substrate. In an electronic instrument where such a type of connector device is used therein, electromagnetic interference that is caused by radiation of an electromagnetic wave or the like may be problematic, so that a shield function may be provided to a connector device.
(3) For example, Japanese Patent Application Publication No. 2008-097849 discloses a connector device that covers a housing where a plurality of signal contacts are arrayed thereon with a shell with a conductive property that is provided on a first connector and a shell with a conductive property that is provided on a second connector. Such a connector device has a configuration for causing a ground member with a conductive property on a first connector to contact a shell with a conducive property on a second connector, and a shield function thereof is improved by such a configuration.
(4) However, further improvement of a shield function in a connector device is desired with a frequency increase of a signal that is transmitted or received, an increase of an operation frequency, or the like, in an electronic instrument where such a connector device is used therein.
SUMMARY OF THE INVENTION
(5) An electrical connector according to an aspect of an embodiment is an electrical connector that is attached to a wiring substrate and mated with a counterpart connector that is connected to a terminal of a signal transmission medium, and that includes a housing that has an insulation property, a plurality of contacts with a conductive property that are arrayed on the housing, and a shell with a conductive property. The shell with a conductive property has a pair of wall parts that are opposed to one another via the plurality of contacts in a direction along a principal surface of the wiring substrate where the direction is orthogonal to an array direction of the plurality of contacts and contact a ground member with a conductive property on the counterpart connector. Among the pair of wall parts, one wall part that is provided at a position that is distant from the signal transmission medium includes a first side plate that is provided with one end that is attached to a ground conductive path of the wiring substrate and extends in a direction away from the wiring substrate from the one end toward another end, a joining part that is provided with one end that is joined to another end of the first side plate, and a second side plate that is provided with one end

that is joined to another end of the joining part and extends in a direction closer to the wiring substrate from the one end toward another end where the other end is a free end. The second side plate has a contact piece that extends in a direction away from the wiring substrate and elastically contacts the ground member of the counterpart connector.

Description

BRIEF DESCRIPTION OF THE DRAWING(S)
(1) FIG. **1** is a diagram illustrating an example of a plug connector and a receptacle connector in a connector device according to an embodiment.
(2) FIG. **2** is a diagram illustrating a state where a plug connector and a receptacle connector according to an embodiment are mated therewith.
(3) FIG. **3** is an appearance perspective view of a plug connector according to an embodiment.
(4) FIG. **4** is an appearance perspective view of a plug connector according to an embodiment.
(5) FIG. **5** is a plan view of a plug connector according to an embodiment.
(6) FIG. **6** is a cross-sectional view along line VI-VI as illustrated in FIG. **5**.
(7) FIG. **7** is a cross-sectional view along line VII-VII as illustrated in FIG. **5**.
(8) FIG. **8** is an appearance perspective view of a receptacle connector according to an embodiment.
(9) FIG. **9** is an appearance perspective view of a receptacle connector according to an embodiment.
(10) FIG. **10** is a plan view of a receptacle connector according to an embodiment.
(11) FIG. **11** is a cross-sectional view along line XI-XI as illustrated in FIG. **10**.
(12) FIG. **12** is a cross-sectional view along line XII-XII as illustrated in FIG. **10**.
(13) FIG. **13** is a plan view of a connector device in a state where a plug connector and a receptacle connector according to an embodiment are mated therewith.
(14) FIG. **14** is a cross-sectional view along line XIV-XIV as illustrated in FIG. **13**.
(15) FIG. **15** is a cross-sectional view along line XV-XV as illustrated in FIG. **13**.
(16) FIG. **16** is a front elevation view of a connector device in a state where a plug connector and a receptacle connector according to an embodiment are mated therewith.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT(S)
(17) Hereinafter, an embodiment(s) of an electrical connector and a connector device as disclosed in the present application will be explained in detail, with reference to the accompanying drawing(s). Additionally, this invention is not limited by an embodiment(s) as illustrated below.
1. Configuration of Connector Device
(18) First, a connector device that includes a plug connector and a receptacle connector according to an embodiment will be explained with reference to FIG. **1** and FIG. **2**.
(19) As illustrated in FIG. **1**, a connector device **1** according to an embodiment includes a plug connector **10** that is connected to terminals of a plurality of coaxial cables **2** and a receptacle connector **20** that is attached to a wiring substrate **3**. The plurality of coaxial cables **2** are an example of a signal transmission medium. Additionally, the plug connector **10** may be a configuration to be connected to a terminal of a signal transmission medium such as a flexible printed circuit (FPC) or a flexible flat cable (FFC), instead of the plurality of coaxial cables **2**. The wiring substrate **3** is, for example, an electrical circuit substrate such as a printed wiring substrate.
(20) The plug connector **10** is an example of a counterpart connector and a first connector and the receptacle connector **20** is an example of an electrical connector and a second connector. Additionally, only a part of the coaxial cables **2** is conveniently illustrated in the drawing(s) and a boundary with a remaining part thereof is illustrated as a cross section thereof.
(21) Furthermore, hereinafter, a direction where the plug connector **10** is inserted into the

receptacle connector **20** therein (a negative direction of a Z-axis) is a “downward direction” and a removal direction (a positive direction of the Z-axis) that is an opposite direction thereof is an “upward direction”, for explanatory convenience. Furthermore, longitudinal directions of the plug connector **10** and the receptacle connector **20** (positive and negative directions of an X-axis) are “leftward and rightward directions”. Furthermore, transverse directions of the plug connector **10** and the receptacle connector **20** (positive and negative directions of a Y-axis) are “frontward and backward directions”.

(22) In the connector device **1**, a plurality of contacts **12** with a conductive property as described later (see FIG. **4**) on the plug connector **10** are electrically connected to a plurality of contacts **22** with a conductive property that are arrayed on a housing **21** of the receptacle connector **20**, by mating between the plug connector **10** and the receptacle connector **20**. Thereby, a center conductor **2***a* as described later (see FIG. **6**) of such a coaxial cable **2** and a non-illustrated signal conductive path that is formed on a principal surface M of the wiring substrate **3** are electrically connected thereto via the contacts **12**, **22**.

(23) As the plug connector **10** is mated with the receptacle connector **20**, the plurality of contacts **12** and the plurality of contacts **22** as described above are covered by a shell **14** with a conductive property on the plug connector **10** and a shell **23** with a conductive property on the receptacle connector **20**. The shells **14**, **23** are ground members that are connected to ground and it is possible to provide the connector device **1** with a shield function by such shells **14**, **23**.

(24) Furthermore, the shell **23** of the receptacle connector **20** has a front wall part **24** and a back wall part **25** that are opposed to one another via the plurality of contacts **22** in frontward and backward directions that are directions along the principal surface M of the wiring substrate **3** where the directions are orthogonal to an array direction of the plurality of contacts **22**. Each of such a front wall part **24** and a back wall part **25** contacts a ground member of the plug connector **10**.

(25) Specifically, the back wall part **25** contacts a conductive member **13** (see FIG. **6**) of the plug connector **10** and the front wall part **24** contacts the shell **14** of the plug connector **10**. Thereby, it is possible to further improve a shield function in the connector device **1**. Additionally, the shell **14** and the conductive member **13** are an example of a ground member. Although the shell **14** and the conductive member **13** are separate bodies, the shell **14** and the conductive member **13** may be formed integrally.

2. Configuration of Plug Connector **10**

(26) Next, a configuration of the plug connector **10** according to an embodiment will be explained with reference to FIG. **3** to FIG. **7**. As illustrated in FIG. **3**, the plug connector **10** is connected to terminals of the plurality of coaxial cables **2**. As illustrated in FIG. **6** and FIG. **7**, the coaxial cable **2** is formed by laminating a dielectric body **2***b*, an outside conductor (a shield line) **2***c* (an example of a first ground conductive path), and an outer circumference covering material **2***d* on an outer circumferential side of a center conductor (a signal line) **2***a* (an example of a first signal conductive path) in order.

(27) As illustrated in FIG. **6** and FIG. **7**, one terminal of the coaxial cable **2** sequentially exposes the center conductor **2***a*, the dielectric body **2***b*, and the outside conductor **2***c* from a distal end side thereof. The outside conductor **2***c* among such exposed parts is sandwiched by a pair of ground bars **4** that are composed of thin plates with a conductive property, so that the plurality of coaxial cables **2** are attached to the ground bars **4**.

(28) As illustrated in FIG. **4**, the plug connector **10** includes a housing **11** with an insulation property (an example of a first housing), the plurality of contacts **12** with a conductive property (an example of a first contact) that are arrayed on the housing **11**, the conductive member **13** that contacts a lower ground bar **4** among the pair of ground bars **4**, and the shell **14** with a conductive property that covers an outer surface of the housing **11** where the plurality of contacts **12** are arrayed thereon and contacts an upper ground bar **4** among the pair of ground bars.

(29) Each of a space between the contacts **12**, a space between such a contact **12** and the conductive member **13**, and a space between the contact **12** and the shell **14** is insulated by the housing **11**. Each of the contacts **12**, the conductive member **13**, and the shell **14** is formed by, for example, applying a punching process or a folding process to a single metal plate.
(30) As illustrated in FIG. **4**, the housing **11** includes a base part **111** with longitudinal directions that are positive and negative directions of an X-axis (leftward and rightward directions), a protrusion part **112** that protrudes from the base part **111** in a positive direction of the X-axis, a protrusion part **113** that protrudes from the base part **111** in a negative direction of the X-axis, and protrusion parts **114**, **115** with an approximately U-shaped cross section where each thereof protrudes downward from the base part **111** and extends in the positive and negative directions of an X-axis (the leftward and rightward directions). As illustrated in FIG. **6**, the protrusion parts **114**, **115** are arranged so as to be spaced apart from one another in frontward and backward directions.
(31) As illustrated in FIG. **4**, the plurality of contacts **12** are arrayed so as to be spaced apart in leftward and rightward directions and fixed on the housing **11** by, for example, insert molding. As illustrated in FIG. **6**, such a plurality of contacts **12** are arranged from the base part **111** to the protrusion part **114** of the housing **11**. Each contact **12** exposes a proximal end part **121** thereof from an upper surface of the base part **111** in the housing **11** and such a proximal end part **121** is connected to the center conductor **2***a* of the coaxial cable **2** by soldering or the like. Furthermore, each contact **12** exposes a distal end part **122** thereof from the protrusion part **114**.
(32) As illustrated in FIG. **4**, the conductive member **13** is arranged on the housing **11** so as to cover a back part of the protrusion part **115**. Such a conductive member **13** is fixed on the housing **11** by, for example, insert molding. As illustrated in FIG. **6**, the conductive member **13** includes a base part **131** that is provided with longitudinal directions that are leftward and rightward directions and contacts such a ground bar **4** and a J-shaped protrusion part **132** that extends downward from the base part **131**. Such a protrusion part **132** is arranged on the protrusion part **115** of the housing **11**.
(33) The shell **14** has a cover upper part **141** that covers an upper part of the housing **11** in a state where one terminal of the coaxial cable **2** is arranged on the plug connector **10** and a cover front part **142** that covers a front part of the housing **11** in a state where the one terminal of the coaxial cable **2** is arranged on the plug connector **10**. As illustrated in FIG. **3**, a plurality of openings **141***a* are formed on the cover upper part **141**. Tongue pieces are formed on edges of such a plurality of openings **141***a* and such a tongue piece and the ground bar **4** are connected by a solder **15**. Additionally, FIG. **3** does not illustrate the solder **15**.
(34) Furthermore, as illustrated in FIG. **3**, the shell **14** has a side protrusion part **143** that protrudes outward from a terminal of the cover upper part **141** in a positive direction of an X-axis and a side protrusion part **144** that protrudes outward from a terminal of the cover upper part **141** in a negative direction of the X-axis. As illustrated in FIG. **4**, a distal end part **143***a* of the side protrusion part **143** is engaged with an engagement recess **112***a* that is formed on the protrusion part **112** of the housing **11** so as to be fixed on the housing **11**. As illustrated in FIG. **3**, a distal end part **144***a* of the side protrusion part **144** is engaged with an engagement recess **113***a* that is formed on the protrusion part **113** of the housing **11** so as to be fixed on the housing **11**.
3. Configuration of Receptacle Connector **20**
(35) Next, a configuration of the receptacle connector **20** according to an embodiment will be explained with reference to FIG. **8** to FIG. **12**. As illustrated in FIG. **8**, the receptacle connector **20** includes a housing **21** with an insulation property (an example of a second housing), a plurality of contacts **22** with a conductive property (an example of a second contact), and a shell **23** with a conductive property that is attached to the housing **21**. Each of the contacts **22** and the shell **23** is formed by, for example, applying a punching process or a folding process to a single metal plate.
(36) As illustrated in FIG. **8** and FIG. **12**, the housing **21** includes a base part **211**, a front wall part **212**, a back wall part **213**, a center wall part **214**, and side wall parts **215**, **216**. The front wall part

212, the back wall part **213**, the center wall part **214**, the side wall part **215**, and the side wall part **216** protrude upward from the base part **211**.

(37) Each of the front wall part **212**, the back wall part **213**, and the center wall part **214** extends in leftward and rightward directions and are arranged so as to be spaced apart from one another in frontward and backward directions. As illustrated in FIG. **8**, the plurality of contacts **22** are arrayed on the center wall part **214** so as to be spaced apart in leftward and rightward directions. As illustrated in FIG. **11**, the plurality of contacts **22** are arranged on the center wall part **214** by press fitting.

(38) As illustrated in FIG. **11** and FIG. **12**, a housing recess **28** where the protrusion part **114** (see FIG. **6**) of the plug connector **10** is housed therein is formed between the front wall part **212** and the center wall part **214**. Furthermore, a housing recess **29** where the protrusion part **115** (see FIG. **6**) of the plug connector **10** is housed therein is formed between the center wall part **214** and the back wall part **213**.

(39) As illustrated in FIG. **11**, such a contact **22** is provided with an intermediate part **222** that is supported by the front wall part **212** of the housing **21** and one terminal **221** that is a free end and is capable of being elastically deformed. The one terminal **221** of the contact **22** is arranged on the center wall part **214** and a contact part **221***a* that contacts the contact **12** of the plug connector **10** is formed on such one terminal **221**. In a state as illustrated in FIG. **11**, the contact part **221***a* protrudes frontward from the center wall part **214** and is arranged on the housing recess **28**. Furthermore, another terminal **223** of the contact **22** is arranged below the front wall part **212** and electrically connected to a non-illustrated signal conductive path (an example of a second signal conductive path) that is formed on the wiring substrate **3** (see FIG. **1**) by soldering or the like.

(40) As illustrated in FIG. **8**, FIG. **9**, and FIG. **11**, the shell **23** includes a front wall part **24** that is arranged so as to be spaced apart from the front wall part **212** in front of the front wall part **212** on the housing **21** and a back wall part **25** that is attached to the back wall part **213** of the housing **21** and covers the back wall part **213**. Furthermore, the shell **23** includes a side wall part **26** that is attached to the side wall part **215** of the housing **21** and covers the side wall part **215** and a side wall part **27** that is attached to the side wall part **216** of the housing **21** and covers the side wall part **216**.

(41) As illustrated in FIG. **9** and FIG. **11**, the front wall part **24** of the shell **23** has a first side plate **31**, a plurality of joining parts **32**, and a second side plate **33**. The first side plate **31** is provided with one end that is attached to a non-illustrated ground conductive path on the wiring substrate **3** and extends in a (upward) direction away from the wiring substrate **3** from the one end toward another end. The plurality of joining parts **32** are provided with ends that are joined to another end of the first side plate **31** and other ends that are joined to one end of the second side plate **33**. The second side plate **33** extends in a (downward) direction closer to the wiring substrate **3** from one end toward another end where the other end is a free end.

(42) As illustrated in FIG. **8**, the first side plate **31** has a ground connection part **311** that is attached to a non-illustrated ground conductive path (an example of a second ground conductive path) on the wiring substrate **3**, an extension part **312** that is continuous with the ground connection part **311** and extends in a (upward) direction away from the wiring substrate **3**, a recess **313** that is a backward dent in a direction toward the back wall part **25** at a position close to one end of the joining part **32**, and a cut part **314**.

(43) As illustrated in FIG. **9** and FIG. **11**, the second side plate **33** has an extension part **331** that is provided with one end that is joined to another side of the joining part **32** and extends in a (downward) direction closer to the wiring substrate **3** from the one end toward another end where the other end is a free end. In an example as illustrated in FIG. **11**, the extension part **331** extends backward and obliquely downward. Additionally, one end of the extension part **331** is one end of the second side plate **33** as described above and another end of the extension part **331** is also another end of the second side plate **33** as described above.

(44) Furthermore, as illustrated in FIG. **8** and FIG. **12**, the second side plate **33** has a plurality of contact pieces **332** that are provided with proximal ends that are joined to the extension part **331** at a position close to another end of the extension part **331** and are cantilevered and supported by the extension part **331**. As illustrated in FIG. **8**, each contact piece **332** is formed into an approximately L-shape and provided with a distal end part that protrudes frontward via the cut part **314**. Furthermore, as illustrated in FIG. **9**, the plurality of contact pieces **332** are plurally formed on the second side plate **33** so as to be spaced apart.
(45) As illustrated in FIG. **12**, such a contact piece **332** has a contact piece extension part **332***a* with one end that is joined to a position close to another end of the extension part **331** on the extension part **331** and a contact part **332***b* that is joined to another end of the contact piece extension part **332***a* and contacts the shell **14** of the plug connector **10**. The contact piece extension part **332***a* extends in a (upward) direction away from the wiring substrate **3** from one end to another end. In an example as illustrated in FIG. **12**, the contact piece extension part **332***a* has a flexural part for forming the contact part **332***b* at a position close to another end, and extends frontward and obliquely upward from a position close to another end of the extension part **331** and subsequently extends frontward and obliquely downward.
(46) As illustrated in FIG. **11**, the contact part **332***b* is arranged to be close to one end (or close to an upper end) of the joining part **32** on the contact piece **332** in upward and downward directions that are directions orthogonal to the principal surface M (see FIG. **1**) of the wiring substrate **3**. Furthermore, the contact part **332***b* is arranged at a position close to the first side plate **31** (or close to an outside) on the second side plate **33** in frontward and backward directions that are directions along the principal surface M of the wiring substrate **3** where the directions are orthogonal to an array direction of the contacts **22**.
(47) As illustrated in FIG. **11**, the back wall part **25** of the shell **23** has a first side plate **41**, a joining part **42**, a second side plate **43**, and a contact piece **44**. The first side plate **41** is provided with one end that is attached to a non-illustrated ground conductive path on the wiring substrate **3** and extends in a (upward) direction away from the wiring substrate **3** from the one end toward another end. The joining part **42** is provided with one end that is joined to another end of the first side plate **41** and another end that is joined to one end of the second side plate **43**. The second side plate **43** extends in a (downward) direction closer to the wiring substrate **3** from one end toward another end.
(48) As illustrated in FIG. **12**, the contact piece **44** is provided with one end that is joined to another end of the joining part **42** and extends in a (downward) direction closer to the wiring substrate **3** from one end toward another end. In an example as illustrated in FIG. **12**, the contact piece **44** extends frontward and obliquely downward. Such a contact piece **44** is cantilevered and supported by the joining part **42** and another end thereof is a free end. For the contact piece **44**, as illustrated in FIG. **12**, a contact part **44***a* that contacts the protrusion part **132** (see FIG. **6**) of the conductive member **13** is arranged on another end of the contact piece **44**. In a state as illustrated in FIG. **11**, the contact piece **44** protrudes frontward from the back wall part **213** and is arranged in the housing recess **29**.
4. Mating State of Plug Connector **10** and Receptacle Connector **20**
(49) Next, a mating state of the plug connector **10** and the receptacle connector **20** will be explained specifically, with reference to FIG. **6** and FIG. **12** to FIG. **15**.
(50) The plug connector **10** is inserted into the receptacle connector **20**, so that the plug connector **10** and the receptacle connector **20** are provided in a mating state thereof. Insertion of the plug connector **10** into the receptacle connector **20** is executed by inserting the protrusion part **114** (see FIG. **6**) of the plug connector **10** into the housing recess **28** (see FIG. **12**) of the receptacle connector **20** and inserting the protrusion part **115** (see FIG. **6**) of the plug connector **10** into the housing recess **29** (see FIG. **12**) of the receptacle connector **20**.
(51) In a case where the plug connector **10** and the receptacle connector **20** are provided in a

mating state thereof, the plurality of contacts **12** with a conductive property on the plug connector **10** are electrically connected to the plurality of contacts **22** with a conductive property on the receptacle connector **20**. Specifically, the distal end part **122** of the contact **12** contacts the contact part **221***a* of the contact **22**, so that the contact **12** and the contact **22** are electrically connected.
(52) The center conductor **2***a* of the coaxial cable **2** is connected to the proximal end part **121** of the contact **12** by soldering or the like, and the other terminal **223** of the contact **22** is connected to a non-illustrated signal conductive path that is formed on the wiring substrate **3**. Hence, in a case where the plug connector **10** and the receptacle connector **20** are provided in a mating state thereof, the center conductor **2***a* of the coaxial cable **2** and a signal conductive path of the wiring substrate **3** are electrically connected.
(53) Furthermore, in a case where the plug connector **10** and the receptacle connector **20** are provided in a mating state thereof, the protrusion part **132** of the conductive member **13** on the plug connector **10** contacts the contact piece **44** of the back wall part **25** on the shell **23**, as illustrated in FIG. **15**. The base part **131** of the conductive member **13** is connected to the outside conductor **2***c* of the coaxial cable **2** via the ground bar **4**, and one end of the first side plate **41** of the back wall part **25** is connected to a non-illustrated ground conductive path that is formed on the wiring substrate **3**, by soldering or the like.
(54) Hence, the outside conductor **2***c* of the coaxial cable **2** is electrically connected to a non-illustrated ground conductive path that is formed on the wiring substrate **3**, via the ground bar **4** and the conductive member **13**. Furthermore, the outside conductor **2***c* of the coaxial cable **2** is electrically connected to the shell **14** of the plug connector **10** via the ground bar **4** and a non-illustrated solder.
(55) Thus, the shell **14** and the shell **23** are electrically connected. The contacts **12**, **22** are surrounded by the shells **14**, **23** that are electrically connected and the conductive member **13**, and thereby, a shield function in the connector device **1** is realized.
(56) Moreover, in a case where the plug connector **10** and the receptacle connector **20** are provided in a mating state thereof, the contact part **332***b* of the contact piece **332** on the second side plate **33** contacts the cover front part **142** of the shell **14**, as illustrated in FIG. **15**. The second side plate **33** is joined to the first side plate **31** via the joining part **32** as illustrated in FIG. **14** and one terminal of the first side plate **31** is connected to a non-illustrated ground conductive path that is formed on the wiring substrate **3**, by soldering or the like.
(57) Hence, the cover front part **142** of the shell **14** is electrically connected to a non-illustrated ground conductive path that is formed on the wiring substrate **3**, via the front wall part **24** of the shell **23**. Thereby, as a conductive path to a non-illustrated ground conductive path that is formed on the wiring substrate **3**, a path that passes through the shell **14** and the front wall part **24** is added to the shell **14** in addition to a path that passes through the conductive member **13** and the back wall part **25**. Hence, it is possible for the connector device **1** to improve a shield function of the shell **14**, so that it is possible to attain further improvement of such a shield function, as compared with a case where the front wall part **24** is absent or a case where the front wall part **24** does not contact the shell **14**.
(58) Furthermore, as illustrated in FIG. **12**, the contact piece **332** is formed on the second side plate **33**, so that a spring length is readily ensured, for example, as compared with a case where it protrudes outward from the joining part **32**. Hence, it is possible to attain downsizing and height reduction of the connector device **1**. Furthermore, one end or a proximal end of the contact piece **332** is joined to a position close to another end (or close to a lower end) of the second side plate **33**, so that a spring length of the contact piece **332** is ensured more readily.
(59) Furthermore, as illustrated in FIG. **12**, the second side plate **33** is arranged between the first side plate **31** and the front wall part **212** of the housing **21**. Hence, as compared with a case where the second side plate **33** is positioned in front of the first side plate **31**, it is possible to reduce a length of the connector device **1** in frontward and backward directions, so that it is possible to

attain downsizing of the connector device **1**.
(60) Furthermore, as illustrated in FIG. **15**, the contact piece **332** has the contact part **332***b* that contacts the shell **14** of the plug connector **10** at a position close to the first side plate **31** in frontward and backward directions. Also thereby, a spring length of the contact piece **332** is ensured more readily.
(61) The contact part **332***b* is arranged to be close to one end (or close to an upper end) of the joining part **32** on the contact piece **332** in upward and downward directions. Thereby, as compared with a case where the contact part **332***b* is arranged to be close to another end (or close to a lower end) of the joining part **32** and the contact part **332***b* contacts the shell **14** at a position close to another end of the joining part **32**, a spring length of the contact piece **332** is ensured more readily. Moreover, as the contact part **332***b* is arranged to be close to one end (or close to an upper end) of the joining part **32** in upward and downward directions, it is also possible to increase a distance of sliding and mating of the contact part **332***b* and the shell **14** (the cover front part **142**) (a so-called effective mating length) as the plug connector **10** and the receptacle connector **20** are mated.
(62) Furthermore, as illustrated in FIG. **14**, the first side plate **31** has the recess **313** at a position close to one end of the joining part **32** in upward and downward directions where the position is opposed to the cover front part **142** of the shell **14** on the plug connector **10**. Such a recess **313** is a backward dent in a direction toward the back wall part **25**.
(63) Hence, as illustrated in FIG. **14**, it is possible to provide a position of the cover front part **142** in frontward and backward directions as a position that is substantially identical to a position of the extension part **312** of the front wall part **24**. Thereby, it is possible to reduce a length of the connector device **1** in frontward and backward directions, so that it is possible to attain downsizing of the connector device **1**.
(64) Meanwhile, in a case where the first side plate **31** is conventionally cut to provide a contact piece, the first side plate **31** has to be cut long from a position close to one end of the first side plate **31** to another end thereof in order to ensure a spring length of such a contact piece. In such a case, a length of the first side plate **31** in upward and downward directions is increased, so that a length of the shell **23** in upward and downward directions is increased. Furthermore, a cut part on the first side plate **31** in upward and downward directions is long, so that an electromagnetic wave leaks from such a cut part and a shield function of the shell **23** is degraded.
(65) The contact piece **332** in the connector device **1** according to the present embodiment is formed on the second side plate **33** that is joined to the first side plate **31** via the joining part **32**. Thereby, the cut part **314** of the first side plate **31** is sufficiently provided with a size not to contact the contact part **332***b* that is present at a position close to one end of the joining part **32** in upward and downward directions.
(66) Hence, as compared with a case where a contact piece is conventionally provided on the first side plate **31**, it is possible to decrease a length of the cut part **314** of the first side plate **31** in upward and downward directions as illustrated in FIG. **16**, so that it is possible to downsize a part of the cur part **314** that is not covered by the cover front part **142** of the shell **14**. Additionally, a size of a part of the cut part **314** that is not covered by the cover front part **142** of the shell **14** is a size not to pass an electromagnetic wave that has a frequency identical to a frequency of a signal that is transmitted by the contacts **12**, **22**, and thereby, it is possible to reduce degradation of a shield function.
(67) Additionally, although the cut part **314** is formed from a position that corresponds to the extension part **312** to a position that corresponds to the joining part **32** in upward and downward directions in an example as illustrated in FIG. **8**, a configuration may be provided in such a manner that it is formed from a position that corresponds to the recess **313** to a position that corresponds to the joining part **32** in upward and downward directions. Thereby, it is possible to cover the cut part **314** with the cover front part **142** of the shell **14**.
(68) Furthermore, as illustrated in FIG. **15**, in a case where the plug connector **10** and the receptacle

connector **20** are provided in a mating state thereof, the protrusion part **132** on the conductive member **13** is positioned in front of the back wall part **25** and the cover front part **142** on the shell **14** is positioned in front of the contact piece **332** on the front wall part **24**. In such a case, the contact piece **332** contacts the cover front part **142** while having a sufficient spring length. Hence, even in a case where a state where the plug connector **10** is tiled relative to the receptacle connector **20** is provided in a process to remove the plug connector **10** from the receptacle connector **20**, it is possible to prevent the contact piece **332** from being deformed plastically.

(69) For example, in a case where the coaxial cable **2** that is connected to the plug connector **10** is gripped and lifted upward, the plug connector **10** is tiled in such a manner that a back part close to the protrusion part **115** is raised while a circumference of the cover front part **142** is a point of support. In such a case, the cover front part **142** of the shell **14** on the plug connector **10** is moved in a direction away from the contact part **332***b* of the contact piece **332**. Hence, it is possible to prevent a great force from acting on the contact piece **332** as compared with a case where the cover front part **142** is positioned in back of the contact piece **332**.

(70) Furthermore, in a case where a front part of the plug connector **10** is lifted in a process to remove the plug connector **10** from the receptacle connector **20**, the plug connector **10** is tiled in such a manner that a front part close to the cover front part **142** is raised while a circumference of the protrusion part **115** is a point of support. In such a case, the cover front part **142** of the shell **14** on the plug connector **10** is moved in a direction away from the contact part **332***b* of the contact piece **332**. Hence, it is possible to prevent a great force from acting on the contact piece **332** as compared with a case where the cover front part **142** is positioned in back of the contact piece **332**.

(71) Furthermore, as illustrated in FIG. **8**, the contact piece **332** is arranged so as to protrude frontward from the cut part **314**. Hence, even in a case where the cover front part **142** of the shell **14** is moved backward, the cover front part **142** contacts the recess **313**, so that it is possible to limit further movement thereof. Hence, also thereby, it is possible to prevent a great force from acting on the contact piece **332**.

(72) As described above, the receptacle connector **20** according to an embodiment is an electrical connector that is attached to the wiring substrate **3** and is mated with the plug connector **10** that is connected to terminals of the plurality of coaxial cables **2**. The plurality of coaxial cables **2** are an example of a signal transmission medium and the plug connector **10** is an example of a counterpart connector. The receptacle connector **20** includes the housing **21** that has an insulation property, the plurality of contacts **22** with a conductive property that are arrayed on the housing **21**, and the shell **23** with a conductive property. The shell **23** has the front wall part **24** and the back wall part **25** that are opposed to one another via the plurality of contacts **22** in a direction along the principal surface M of the wiring substrate **3** where the direction is orthogonal to an array direction of the plurality of contacts **22**, and contact the conductive member **13** and the shell **14** that are provided with a conductive property on the plug connector **10**. The front wall part **24** and the back wall part **25** are an example of a pair of wall parts. The front wall part **24** is an example of one wall part. The conductive member **13** and the shell **14** are an example of a ground member. The front wall part **24** includes the first side plate **31** that is provided with one end that is attached to a ground conductive path of the wiring substrate **3** and extends in a direction away from the wiring substrate **3** from the one end toward another end, the joining part **32** that is provided with one end that is joined to another end of the first side plate **31**, and the second side plate **33** that is provided with one end that is joined to another end of the joining part **32** and extends in a direction closer to the wiring substrate **3** from the one end toward another end where the other end is a free end. Then, the second side plate **33** has the contact piece **332** that extends in a direction away from the wiring substrate **3** and elastically contacts the shell **14** of the plug connector **10**. Thereby, it is possible for the receptacle connector **20** according to an embodiment to attain further improvement of a shield function.

(73) Furthermore, the second side plate **33** has the extension part **331** that is provided with one end

that is joined to another end of the joining part **32** and extends in a direction closer to the wiring substrate **3** from the one end toward another end where the other end is a free end. The contact piece **332** is provided with one end that is joined to a position close to another end of the extension part **331** on the extension part **331** and extends in a direction away from the wiring substrate **3** from the one end toward another end. Thereby, it is possible to readily ensure a spring length, for example, as compared with a case where it protrudes from the joining part **32**.
(74) Furthermore, the second side plate **33** is arranged between the first side plate **31** and the housing **21**. It is possible to reduce a length of the connector device **1** in frontward and backward directions as compared with a case where the second side plate **33** is positioned in front of the first side plate **31**, so that it is possible to attain downsizing of the connector device **1**.
(75) Furthermore, the contact piece **332** has the contact part **332***b* that contacts the shell **14** of the plug connector **10** at a position close to the first side plate **31** in a direction along the principal surface M of the wiring substrate **3** where the direction is orthogonal to an array direction of the contacts **22**. Thereby, it is possible to ensure a spring length of the contact piece **332** more readily.
(76) Furthermore, the contact part **332***b* is arranged to be close to one end of the joining part **32** on the contact piece **332** in a direction orthogonal to the principal surface M of the wiring substrate **3**. Thereby, a spring length of the contact piece **332** is ensured more readily.
(77) Furthermore, the first side plate **31** has the recess **313** at a position close to one end of the joining part **32** in a direction orthogonal to the principal surface M of the wiring substrate **3** where the position is opposed to the cover front part **142** of the shell **14** of the plug connector **10**, and such a recess **313** is a dent in a direction toward the back wall part **25**. Thereby, it is possible to attain downsizing of the connector device **1**. The back wall part **25** is an example of another wall part.
(78) According to an aspect of an embodiment, it is possible to provide an electrical connector and a connector device that are capable of attaining further improvement of a shield function thereof.
(79) An additional effect or variation can readily be derived by a person(s) skilled in the art. Hence, a broader aspect of the present invention is not limited to a specific detail(s) and a representative embodiment(s) as illustrated and described above. Therefore, various modifications are allowed without departing from a spirit and a scope of a general inventive concept as defined by the appended claim(s) and an equivalent(s) thereof.

Claims

1. An electrical connector, wherein the electrical connector is mated with a counterpart connector that is attached to a principal surface of a wiring substrate, wherein the counterpart connector includes: a first housing that has an insulation property; a plurality of first contacts with a conductive property that are arrayed on the first housing; and a pair of wall parts with a conductive property that are opposed to one another through the plurality of first contacts in a direction that is a direction along the principal surface and is orthogonal to an array direction of the plurality of first contacts, one wall part in the pair of wall parts with a conductive property is provided with: a cut part; and a contact piece with a conductive property that is provided with a part that protrudes through the cut part, the one wall part is located outside end parts of the plurality of first contacts that are electrically connected to a conductive path of the wiring substrate in a direction that is a direction along the principal surface and is orthogonal to the array direction, the electrical connector includes: a second housing that has an insulation property; a plurality of second contacts with a conductive property that are arrayed on the second housing in a first direction that is provided as an array direction thereof and are connected to the plurality of first contacts; a first shell with a conductive property that has a first cover part that is opposed to the plurality of second contacts in a second direction that is orthogonal to the first direction and covers the plurality of second contacts and a second cover part that extends in the second direction from one end part of the first cover part in a third direction that is orthogonal to the first direction and the second

direction; and a second shell with a conductive property that is arranged near another end part of the first cover part in the third direction, in a state where the counterpart connector and the electrical connector are mated, the first cover part is opposed to the plurality of first contacts in the second direction and covers a whole of the plurality of first contacts, the second cover part covers a part or all of the cut part in a state where it contacts the contact piece, and the second shell is opposed to and contacts another wall part in the pair of wall parts with a conductive property in the third direction, and in a state where the electrical connector and the counterpart connector are mated, the electrical connector has a housing space that houses a part of the one wall part near the first cover part to have a gap with a part of the part of the one wall part near the another wall part.

2. An electrical connector, wherein the electrical connector is attached to a principal surface of a wiring substrate and is mated with a counterpart connector, wherein the electrical connector includes: a first housing that has an insulation property; a plurality of first contacts with a conductive property that are arrayed on the first housing; and a pair of wall parts with a conductive property that are opposed to one another through the plurality of first contacts in a direction that is a direction along the principal surface and is orthogonal to an array direction of the plurality of first contacts, one wall part in the pair of wall parts with a conductive property is provided with: a cut part; and a contact piece with a conductive property that is provided with a part that protrudes through the cut part, the one wall part is located outside end parts of the plurality of first contacts that are electrically connected to a conductive path of the wiring substrate in a direction that is a direction along the principal surface and is orthogonal to the array direction, the counterpart connector includes: a second housing that has an insulation property; a plurality of second contacts with a conductive property that are arrayed on the second housing in a first direction that is provided as an array direction thereof and are connected to the plurality of first contacts; a first shell with a conductive property that has a first cover part that is opposed to the plurality of second contacts in a second direction that is orthogonal to the first direction and covers the plurality of second contacts and a second cover part that extends in the second direction from one end part of the first cover part in a third direction that is orthogonal to the first direction and the second direction; and a second shell with a conductive property that is arranged near another end part of the first cover part in the third direction, in a state where the electrical connector and the counterpart connector are mated, the first cover part is opposed to the plurality of first contacts in the second direction and covers a whole of the plurality of first contacts, the second cover part covers a part or all of the cut part in a state where it contacts the contact piece, and the second shell is opposed to and contacts another wall part in the pair of wall parts with a conductive property in the third direction, and in a state where the electrical connector and the counterpart connector are mated, the counterpart connector has a housing space that houses a part of the one wall part near the first cover part to have a gap with a part of the part of the one wall part near the another wall part.

3. An electrical connector, wherein the electrical connector is mated with a counterpart connector that is attached to a principal surface of a wiring substrate, wherein the counterpart connector includes: a first housing that has an insulation property; a plurality of first contacts with a conductive property that are arrayed on the first housing; and a pair of wall parts with a conductive property that are opposed to one another through the plurality of first contacts in a direction that is a direction along the principal surface and is orthogonal to an array direction of the plurality of first contacts, one wall part in the pair of wall parts with a conductive property is provided with: a cut part; and a contact piece with a conductive property that is provided with a part that protrudes through the cut part, the one wall part is located outside end parts of the plurality of first contacts that are electrically connected to a conductive path of the wiring substrate in a direction that is a direction along the principal surface and is orthogonal to the array direction, the electrical connector includes: a second housing that has an insulation property; a plurality of second contacts with a conductive property that are arrayed on the second housing in a first direction that is

provided as an array direction thereof and are connected to the plurality of first contacts; a first shell with a conductive property that has a first cover part that is opposed to the plurality of second contacts in a second direction that is orthogonal to the first direction and covers the plurality of second contacts and a second cover part that extends in the second direction from one end part of the first cover part in a third direction that is orthogonal to the first direction and the second direction; and a second shell with a conductive property that is arranged near another end part of the first cover part in the third direction, in a state where the counterpart connector and the electrical connector are mated, the first cover part is opposed to the plurality of first contacts in the second direction and covers a whole of the plurality of first contacts, the second cover part covers a part or all of the cut part in a state where it contacts the contact piece, and the second shell is opposed to and contacts another wall part in the pair of wall parts with a conductive property in the third direction, a part of the one wall part near the first cover part has a recess that is a dent in a direction toward another wall part in the pair of wall parts, and in a state where the electrical connector and the counterpart connector are mated, the electrical connector has a housing space that houses the part of the one wall part near the first cover part.

4. An electrical connector, wherein the electrical connector is attached to a principal surface of a wiring substrate and is mated with a counterpart connector, wherein the electrical connector includes: a first housing that has an insulation property; a plurality of first contacts with a conductive property that are arrayed on the first housing; and a pair of wall parts with a conductive property that are opposed to one another through the plurality of first contacts in a direction that is a direction along the principal surface and is orthogonal to an array direction of the plurality of first contacts, one wall part in the pair of wall parts with a conductive property is provided with: a cut part; and a contact piece with a conductive property that is provided with a part that protrudes through the cut part, the one wall part is located outside end parts of the plurality of first contacts that are electrically connected to a conductive path of the wiring substrate in a direction that is a direction along the principal surface and is orthogonal to the array direction, the counterpart connector includes: a second housing that has an insulation property; a plurality of second contacts with a conductive property that are arrayed on the second housing in a first direction that is provided as an array direction thereof and are connected to the plurality of first contacts; a first shell with a conductive property that has a first cover part that is opposed to the plurality of second contacts in a second direction that is orthogonal to the first direction and covers the plurality of second contacts and a second cover part that extends in the second direction from one end part of the first cover part in a third direction that is orthogonal to the first direction and the second direction; and a second shell with a conductive property that is arranged near another end part of the first cover part in the third direction, in a state where the electrical connector and the counterpart connector are mated, the first cover part is opposed to the plurality of first contacts in the second direction and covers a whole of the plurality of first contacts, the second cover part covers a part or all of the cut part in a state where it contacts the contact piece, and the second shell is opposed to and contacts another wall part in the pair of wall parts with a conductive property in the third direction, a part of the one wall part near the first cover part has a recess that is a dent in a direction toward another wall part in the pair of wall parts, and in a state where the electrical connector and the counterpart connector are mated, the counterpart connector has a housing space that houses the part of the one wall part near the first cover part.